



DESeq2 moderated dispersion and fold-change estimation

A paper talk on negative-binomial RNA-seq inference

Research Presentation Program

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DESeq2 starts from a negative-binomial GLM

$$K_{ij} \sim NB(\mu_{ij}, \alpha_i), \quad \mu_{ij} = s_j q_{ij}, \quad \log_2 q_{ij} = x_j \cdot \beta_i$$

The GLM frame lets DESeq2 estimate expression effects while modeling mean-dependent count variance.

Model components

- negative binomial counts
- size factors
- dispersion α_i

Interpretation

The model separates sequencing-depth normalization, expected expression, and gene-specific dispersion.

Love, Huber, and Anders 2014, Methods.

Fold-change shrinkage stabilizes weakly informed effects

Love et al. *Genome Biology* 2014 15:1050

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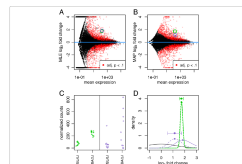


Figure 2 Effect of shrinkage on log2 fold change estimates. Panel A: **DESeq2** (unshrunken) and **limma** (shrunken) estimates are shown. Panel B: **DESeq2** (unshrunken) and **limma** (shrunken) estimates are shown. Panel C: **DESeq2** (unshrunken) and **limma** (shrunken) estimates are shown. Panel D: **DESeq2** (unshrunken) and **limma** (shrunken) estimates are shown.

Shrinkage depends on information: low counts, high dispersion, or few degrees of freedom pull LFC estimates more strongly toward zero.

Figure 2 page, Love et al. 2014.



Dispersion shrinkage borrows strength across genes

Love et al. *Genome Biology* (2014) 15:550

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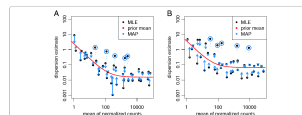


Figure 1 Shrinkage estimation of dispersion. This of dispersion estimates use the average expression strength (A) to the Ratanjy et al. [24] dataset with an average across two genes and (B) for an average from the Ratanjy et al. [24] dataset. The only prior dispersion (prior mean) MLEs are obtained using only the respective gene's data (black dots). Then a curve (red) is fit to the MLEs to capture the overall trend of dispersion versus expression. This fit is used as a prior mean for a second dispersion model, which results in the final MAP estimate of dispersion (blue dots). This can be understood as a shrinkage (toward the fit) of the noisy gene-wise estimates toward the consensus represented by the red line. The blue points (circles) that are observed as dispersion outliers and are shrunk toward the prior (shrinkage) result (blue dots) are shown. For clarity only a subset of genes is shown, which is enriched for dispersion outliers. Additional Fig. 1 Fig. S1 displays the same data but with dispersion of all genes shown. MAP, maximum a posteriori; MLE, maximum likelihood estimate.

variation to the extent that the data provide this information, while the fitted curve aids estimation and testing in low information-rich settings.

Our approach is similar to the one used by DESeq [1], in that both methods sequentially estimate a prior distribution for the true dispersion values across the fit, and then provide the maximum a posteriori (MAP) as the final estimate. It differs from the previous implementation of DESeq, which used the maximum of the fitted curve and the gene-wise dispersion estimate as the final estimate and tended to overestimate the dispersions (Additional Fig. 1 Figure S2). The approach of DESeq2 differs from that of edgeR [1], as DESeq2 estimates the width of the prior distribution from the data and therefore automatically controls the amount of shrinkage based on the observed properties of the data. In contrast, the default recipe in edgeR requires a user-specified parameter, the prior degrees of freedom, which weighs the contribution of the individual gene estimate and edgeR's dispersion fit.

Note that in Figure 1, a number of genes with gene-wise dispersion estimates below the curve have their final estimates raised substantially. The shrinkage procedure thereby helps avoid potential false positives, which can result from underestimates of dispersion. It can, on the other hand, as individual genes' dispersion is far above the distribution of the gene-wise dispersion estimates of other genes, then the shrinkage would tend to a greatly reduced final estimate of dispersion. We reasoned that in many cases, the reason for extraordinarily high dispersion of a

gene is that it does not obey our modeling assumptions: some genes may show much higher variability than others for biological or technical reasons, even though they have the same average expression levels. In these cases, inference based on the deviation dispersion estimates could lead to undesirable false positive calls. DESeq2 handles these cases by using the gene-wise estimate instead of the deviation estimate when the former is more than 2 residual standard deviations above the curve.

Empirical Bayes shrinkage for fold-change estimation. A common difficulty in the analysis of HTS data is the strong variance of LFC estimates for genes with low read count. We demonstrate this issue using the dataset by Ratanjy et al. [18]. As visualized in Figure 2A, weakly expressed genes seem to show much stronger differences between the compared mouse strains than strongly expressed genes. This phenomenon, seen in most HTS datasets, is a direct consequence of dealing with count data, in which ratios are inherently noisier when counts are low. This heteroskedasticity (variance of LFCs depending on mean count) complicates downstream analysis and data interpretation, as it makes effect sizes difficult to compare across the dynamic range of the data.

DESeq2 overcomes this issue by shrinking LFC estimates toward zero in a manner such that shrinkage is stronger when the available information for a gene is low, which may be because counts are low, dispersion is high or there are few degrees of freedom. We again employ an empirical Bayes procedure: we first perform

Figure 1 page 3 of shrinkage 2014 Figure BY is the core mechanism: noisy estimates move toward a fitted consensus unless treated as outliers.

Benchmark gains are empirical, not a license to ignore design

Love et al. *Genome Biology* (2014) 15:250

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procedure for differential expression analysis described above, which employs the raw counts, not transformed data.

The rlog transformation is calculated by fitting for each gene a GLM with a baseline expression (i.e. intercept) and, computing for each sample, standard LFCs with respect to this baseline, using the same statistical test procedure as before (Wald). Then, however, the sample covariance information (i.e. dispersion) is omitted so we seek to find all samples are treated equally. The rlog transformation accounts for variation in sequencing depth across samples as it represents the logarithm of y_i after accounting for the size factors s_i . This is in contrast to the variance-stabilizing transformation (VST) for overdispersed counts introduced in DESeq (3), while the VST is also effective at stabilizing variance, it does not directly take into account differences in size factors, and in datasets with large variation in sequencing depth (differences in total count factors) (34), we observed undesirable artifacts in the performance of the VST. A disadvantage of the log

transformation with respect to the VST is, however, that the ordering of genes within a sample will change if multiple genes undergo fold-changes of different strength. As with the VST, the value of $\log_2(x_i/s_i)$ for large counts is approximately equal to $\log_2(x_i/s_i)$, both the log transformation and the VST are provided in the `DESeq2` package.

We demonstrate the use of the log transformation on the RNA-seq dataset of Hansen et al. (24), where there was separation for the control and sample-to-rate that had unique spatial source locations and counts of 0.7 counts and 0.2 counts after the log transformation. The count matrix for this dataset was downloaded from the Bioconductor dataset (35). This dataset offers more subtle differences between conditions than the Ikonidis et al. (33) dataset. Figure 5 provides diagnostic plots of this normalized counts under the ordinary together with a pseudocount of 1 and for the log transformation, showing that the log transformation stabilizes the variance across the range of the mean of counts and helps to find meaningful patterns in the data.

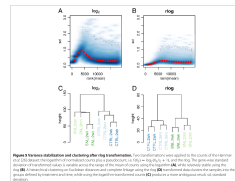


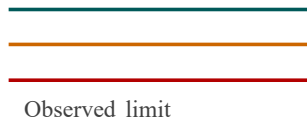
Figure 5 Variance stabilization and checking after log transformation. Variance stabilization was applied to the counts of the Hansen et al. (24) RNA-seq dataset and the results are shown in panels A and B. The density of normalized counts is visible across the range of the mean of counts across the log2(counts) (A) and the log2(counts) + 1 (B). The dendrograms show the distribution of normalized counts is visible across the range of the mean of counts across the log2(counts) (C) and the log2(counts) + 1 (D). The dendrograms show that the log transformation stabilizes the variance across the range of the mean of counts and helps to find meaningful patterns in the data.

Figure 6 supports comparative performance claims, but the paper still frames testing around design, FDR, and effect-size thresholds. Figure 6 page, Love et al. 2014.

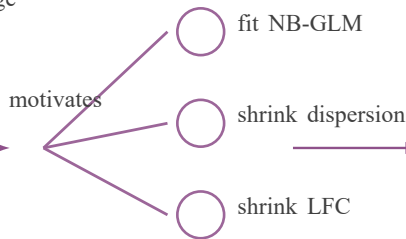
The durable idea is moderated estimation, not more filters

Failure evidence

Ranking genes by raw fold change can overstate weakly supported effects in low-count data.

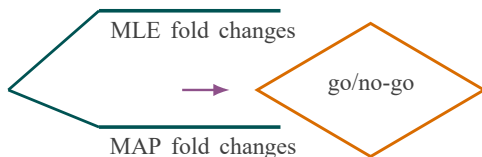


Manipulate sampling



Comparator arms

Decision rule



Use DESeq2 when the scientific question benefits from moderated dispersion and effect-size estimates with transparent design assumptions.

The paper-specific close is an estimation contract: regularize noisy gene-wise quantities before ranking or testing.



The comparison object is sensitivity versus precision

Love et al. *Bioinformatics* (2014) 30:1360

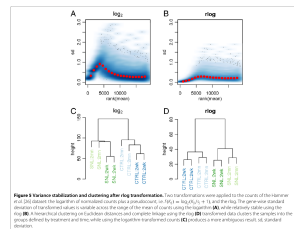
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procedure for differential expression analysis described above, which employs the raw counts, not transformed data.

The *ring* transformation is calculated by fitting for each gene a GLM with a baseline expression (i.e., intercept only) and, computing for each sample, shrunken LFCs with respect to the baseline, using the same empirical Bayes procedure as before (Methods). Here, however, the sample covariate information (e.g., treatment or control) is not used, so that all samples are treated equally. The *ring* transformation accounts for variation in sequencing depth across samples as it represents the logarithm of ϕ_j after accounting for the size factors s_j . This is in contrast to the variance-stabilizing transformation (VST) for overdispersed counts, introduced in DESeq (4), while the VST is also effective at stabilizing variance, it does not directly take into account differences in size factors, and is datasets with large variation in sequencing depth (dynamic range of size factors ≥ 4) we observed undesirable artifacts in the performance of the VST. A disadvantage of the *ring*

transformation with respect to the VST is, however, that the ordering of genes within a sample will change if neighbouring genes undergo shrinkage of different strength. As with the VST, the value of $\log_2(\phi_j)$ for large counts is approximately equal to $\log_2(\mu_j/s_j)$. Both the *ring* transformation and the VST are provided in the DESeq2 package.

We demonstrate the use of the *ring* transformation on the RNA-seq dataset of Hammer et al. (36), wherein RNA was sequenced from the distal root apical meristems of rats that had undergone spinal nerve ligation and controls, at 3 weeks and at 3 months after the ligation. The count matrix for this dataset was downloaded from the RefGene data resource (37). This dataset offers more subtle differences between conditions than the Retortier et al. (34) dataset. Figure 5 provides diagnostic plots of the normalized counts under the ordinary log-linear with a pseudocount of 1 and the *ring* transformation, showing that the *ring* both stabilizes the variance through the range of the means of counts and helps to find meaningful patterns in the data.



This comparison is not relevant for DESeq2 benchmark page: it is about differential-expression algorithms, not a generic omics pipeline